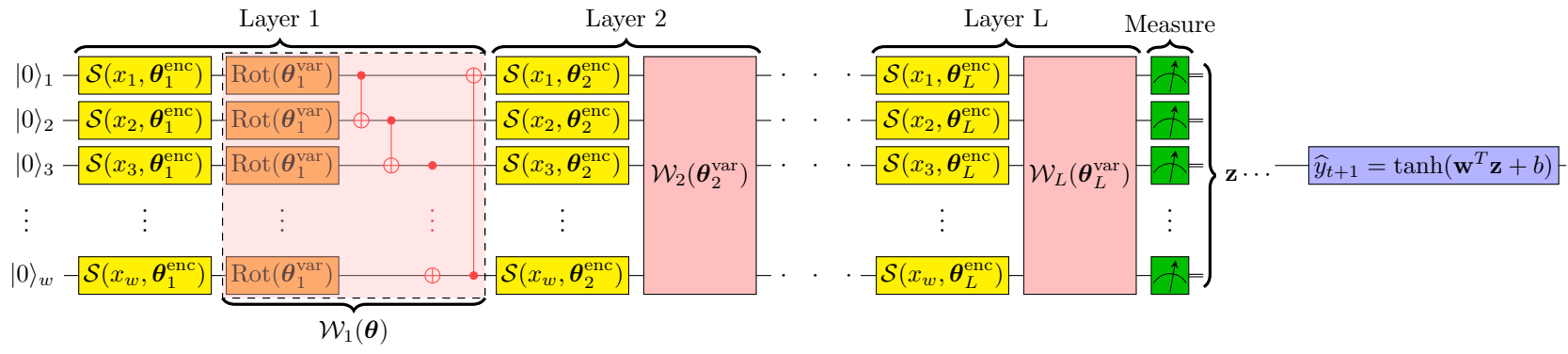

Layer 1
Layer 2
Layer L
Measure
$|0\rangle_1$
$|0\rangle_2$
$|0\rangle_3$
$|0\rangle_w$
$\mathcal{S}(x_1, \boldsymbol{\theta}_1^{\text{enc}})$
$\mathcal{S}(x_2, \boldsymbol{\theta}_1^{\text{enc}})$
$\mathcal{S}(x_3, \boldsymbol{\theta}_1^{\text{enc}})$
$\mathcal{S}(x_w, \boldsymbol{\theta}_1^{\text{enc}})$
$\text{Rot}(\boldsymbol{\theta}_1^{\text{var}})$
$\mathcal{W}_1(\boldsymbol{\theta})$
$\mathcal{S}(x_1, \boldsymbol{\theta}_2^{\text{enc}})$
$\mathcal{S}(x_2, \boldsymbol{\theta}_2^{\text{enc}})$
$\mathcal{S}(x_3, \boldsymbol{\theta}_2^{\text{enc}})$
$\mathcal{S}(x_w, \boldsymbol{\theta}_2^{\text{enc}})$
$\mathcal{W}_2(\boldsymbol{\theta}_2^{\text{var}})$
$\mathcal{S}(x_1, \boldsymbol{\theta}_L^{\text{enc}})$
$\mathcal{S}(x_2, \boldsymbol{\theta}_L^{\text{enc}})$
$\mathcal{S}(x_3, \boldsymbol{\theta}_L^{\text{enc}})$
$\mathcal{S}(x_w, \boldsymbol{\theta}_L^{\text{enc}})$
$\mathcal{W}_L(\boldsymbol{\theta}_L^{\text{var}})$
$\mathbf{z}$
$\widehat{y}_{t+1} = \tanh(\mathbf{w}^T\mathbf{z} + b)$